

THAMES TOWER, United Kingdom

WMO: 036950

Lat: 51.667N

Lon: 1.100E

Elev: 8

StdP: 101.23

Time Zone: 0.00 (EUW)

Period: 82-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.1	0.7	-4.6	2.6	0.6	-2.6	3.0	2.3	20.4	3.8	19.3	3.9	10.4	70	0.746

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.5	21.7	19.3	20.9	18.8	20.1	18.2	20.0	21.0	19.3	20.4	18.6	19.8	5.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	14.3	20.8	18.8	13.7	20.1	18.1	13.0	19.6	57.3	21.2	54.9	20.4	52.8	19.9	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.8	14.2	12.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	5.8	5.4	7.0	8.5	11.6	14.4	17.2	17.8	15.4	12.7	8.9	7.4
	DBStd	4.83	2.70	2.80	2.16	2.14	2.22	1.97	1.72	1.90	1.83	2.33	2.75	2.69
	HDD10.0	566	132	128	94	55	11	0	0	0	0	6	52	88
	HDD18.3	2687	389	361	351	297	210	117	43	33	89	175	282	339
	CDD10.0	948	1	1	2	9	59	133	224	242	161	89	20	7
	CDD18.3	27	0	0	0	0	0	1	9	17	1	0	0	0
	CDH23.3	2	0	0	0	0	0	0	1	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	6.8	7.9	7.3	6.4	6.5	6.5	6.2	5.8	6.3	6.8	7.4	6.7	7.5
(15) Precipitation	PrecAvg	567	50	40	34	37	42	50	45	53	47	60	58	53
	PrecMax	749	107	93	90	98	96	121	93	114	98	148	117	105
	PrecMin	388	11	5	9	0	8	10	9	5	8	1	19	12
	PrecStd	93	23	24	21	26	26	33	20	32	26	33	27	27
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	11.6	12.7	14.4	17.2	20.4	22.8	23.2	20.7	18.3	15.5	13.7
		MCWB	10.8	10.3	11.1	12.6	15.2	18.1	19.2	20.4	17.5	16.4	14.7	13.1
	2%	DB	10.7	10.6	11.6	13.1	16.1	19.0	21.5	21.9	19.5	17.1	14.5	13.0
		MCWB	9.7	9.5	10.2	11.8	14.4	17.0	18.9	19.6	16.9	15.5	13.7	12.6
	5%	DB	9.9	9.9	10.8	12.1	15.3	18.0	20.6	21.2	18.8	16.2	13.5	12.2
		MCWB	9.0	8.7	9.5	10.9	13.8	16.4	18.3	19.1	16.5	14.8	12.7	11.7
	10%	DB	9.2	9.1	10.0	11.4	14.6	17.2	19.8	20.6	18.0	15.6	12.7	11.3
		MCWB	8.3	8.0	8.9	10.3	13.2	15.7	17.8	18.7	16.1	14.2	11.7	10.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	10.8	11.4	13.2	15.7	18.5	20.7	21.3	18.3	16.9	15.1	13.6
		MCDB	11.8	11.3	12.3	14.0	16.8	19.7	21.5	22.4	19.3	17.6	15.3	13.7
	2%	WB	10.0	9.7	10.5	12.0	14.7	17.4	19.7	20.4	17.3	15.9	14.0	12.8
		MCDB	10.6	10.4	11.3	12.7	15.7	18.4	20.6	21.5	18.3	16.6	14.4	13.0
	5%	WB	9.2	9.1	9.8	11.2	14.1	16.6	18.8	19.7	16.8	15.2	13.0	11.9
		MCDB	9.8	9.8	10.6	11.9	15.1	17.6	19.8	20.8	17.8	16.1	13.4	12.2
	10%	WB	8.6	8.3	9.2	10.5	13.5	16.0	18.1	19.2	16.2	14.5	11.8	10.9
		MCDB	9.2	9.1	9.9	11.3	14.5	17.0	19.4	20.3	17.2	15.5	12.5	11.3
(35) Mean Daily Temperature Range	5% DB	MDBR	2.9	2.7	2.8	3.0	2.9	3.3	3.5	3.5	3.2	3.0	2.7	2.8
		MCDBR	3.9	3.4	3.7	3.8	3.6	4.4	4.4	4.2	3.6	3.3	2.4	3.2
	5% WB	MCWBR	4.0	3.2	3.4	3.1	2.8	3.4	3.3	3.3	2.8	2.8	2.3	3.1
		MCWBR	3.7	3.3	3.7	3.6	3.4	4.0	3.8	3.5	3.2	2.9	2.3	2.8
(40) Clear-Sky Solar Irradiance	taub		0.331	0.348	0.384	0.405	0.412	0.414	0.411	0.403	0.382	0.365	0.338	0.323
		taud	2.381	2.386	2.280	2.254	2.282	2.301	2.318	2.364	2.412	2.433	2.421	2.373
	Ebn at Noon		671	768	802	830	843	843	838	826	801	739	659	614
		Edh at Noon	62	80	107	123	125	124	120	109	93	76	59	54
(44) All-Sky Solar Radiation	RadAvg		0.77	1.44	2.56	4.10	4.89	5.39	5.13	4.28	3.21	1.84	0.94	0.60
	RadStd		0.07	0.22	0.31	0.39	0.35	0.42	0.47	0.33	0.26	0.17	0.11	0.08

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (80 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page